Venkat Pabbathi *UI/UX Designer & Full-Stack Developer*

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PROFILE

UI/UX designer with a background in full-stack development (React, Next.js, Node.js, MongoDB). Currently designing the UI for an AI-based data-mapping tool and dashboards for an energy-optimisation research project at the August-Wilhelm Scheer Institut. In parallel, designing and developing a web-based decision-support interface and running a controlled user study for a Master's thesis at Fraunhofer IPA. MSc student in Mathematics and Computer Science at Saarland University, focusing on human-computer interaction. I design screens with the stack in mind, for the happy path and the broken one. Looking for roles that combine UI/UX design with frontend development.

SKILLS

Programming — JavaScript, TypeScript, Python

Frontend — React, Next.js, HTML5, CSS3

Backend — Node.js, REST APIs

Databases — MongoDB, MySQL

Tools — Git, GitHub, Linux CLI, AWS, Figma

UI/UX — User Research, Wireframing & Prototyping, Interaction Design, Usability Evaluation, Design Systems, Accessibility (WCAG), Responsive Design

PROFESSIONAL EXPERIENCE

Junior Researcher (Werkstudent), UI/UX Design 07/2026 – Present | Saarbrücken, Germany
August-Wilhelm Scheer Institut – Smart Energy Lab

- Created and iteratively refined **Figma wireframes and prototypes** for **AutoMap.AI**, an AI-based data-mapping tool (5-step flow, dashboard, format library)
- Bridged **interface and backend**: reviewed backend code and tested the running app to learn what the backend can deliver, shaping confidence levels, review/retry flows and error states
- Built and documented a **design system** (components, type scale, spacing), applied **WCAG accessibility guidelines**, and adjusted layouts after **frontend feedback**; parts are implemented by developers
- Turning requirements and mapped data flows into **role-specific dashboards** for **EnOptI**, an energy-optimisation research project

Full-Stack Developer 08/2022 – 03/2023 | Bengaluru, India
Amagi Media Labs

- Built and maintained full-stack features using **React.js, Node.js, MongoDB, and MySQL** to support internal media content management workflows.
- Developed **responsive dashboard components** in **React.js** to monitor and manage media assets and delivery pipelines.
- Designed and integrated **REST APIs** for structured media asset retrieval and content distribution.
- Deployed and maintained cloud services using **AWS (S3, EC2)** for storage, processing, and monitoring.

Full-Stack Development Intern 12/2021 – 05/2022 | Pune, India
Scalefull Technologies

- Built and maintained web application features using **React, Node.js, Python, MongoDB, and MySQL** across frontend and backend components
- Engineered **responsive UI components** and supported backend functionality to improve application usability and performance
- Developed **REST-based services** and database interactions to support scalable web application workflows
- Collaborated with senior engineers in an **Agile** development environment to deliver reliable and maintainable features

MASTER'S THESIS

Working title: Interface Design for Pairwise Comparison in a Multi-Criteria Decision-Support Tool 07/2026 – Present | Stuttgart, Germany

Fraunhofer IPA

- **Designing and developing** a **web-based interface** to replace an **Excel-based workflow** in a multi-criteria decision-support tool
- Analysed the existing **Excel/Python pipeline** (utility curves, **AHP pairwise-comparison** weighting, **Monte Carlo simulation**) to define requirements for the interface
- **Planning and conducting** requirements interviews and **think-aloud usability evaluation** with Fraunhofer researchers
- Designing a **controlled user study** comparing three ways of displaying value information during pairwise comparison

PROJECTS

Anilog – Full-Stack Anime Tracker 04/2022 – 06/2022

- Built a full-stack anime tracking platform using React and Next.js, focusing on **responsive UI/UX design** and efficient state management to handle large anime libraries.
- Designed and implemented **REST APIs** for structured **content retrieval, filtering, and user-specific data management** between frontend and database layers.
- Implemented **authentication, personalized watchlists, and dynamic filtering**, allowing users to track and manage viewing progress efficiently.
- Deployed the application on **Vercel / Render** with environment-based configuration and **secure session handling.**

Live: https://anilog-tracker.vercel.app/

Dual-Gyro Arduino Coach 04/2025 – 06/2025

- Developed a **real-time motion monitoring system** using **Arduino and gyroscope sensors** to detect posture and movement patterns.
- Processed live sensor data using **time-based threshold detection** to identify posture deviations and rhythmic motion.
- Implemented **serial data logging** and **audio-visual feedback mechanisms** for immediate user feedback.

IKnow Study App 10/2024 – 01/2025

- Designed and prototyped a **productivity-focused study application** using **Figma**, specifically tailored to the needs of university students.
- Conducted comprehensive **UX research**, including developing user personas, task analysis, and **usability testing** to identify core user pain points.
- Translated research insights into **structured interaction flows** and high-fidelity prototypes, focusing on **web accessibility** and **reducing cognitive load** for better retention.
- Applied **HCI principles** from my Master's studies to ensure the interface was intuitive, minimizing user friction during high-intensity study sessions.

Figma Prototype: IKnow

EDUCATION

MSc — Mathematics and Computer Science 04/2023 – Present | Saarbrücken, Germany

Universität des Saarlandes

Expected Graduation: September 2027
Focus: **Human-Computer Interaction, Interactive Systems, and Software Engineering (ongoing)**

BSc — Mathematics, Computer Science, Electronics 08/2019 – 07/2022 | Bengaluru, India

St. Joseph College (Autonomous)

- Core foundation in **Data Structures, Algorithms, Database Systems, and Object-Oriented Programming**
- Bachelor Project: Developed a web-based clinic management system using **PHP and MySQL**

LANGUAGES

German — Working Proficiency

English — Fluent